

Analyzing Crude Oil Price Movements with Predictive Machine Learning Techniques

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Abstract— Prices of crude oil are renowned for being volatile, and they are governed by a broad set of factors including global economic scenarios, political tensions, and supply-demand ratios. Precise forecasting of the prices of crude oil is pivotal for investors, policymakers, as well as for business players, so that appropriate choices can be made. Here, we propose a hybrid machine learning (ML) model to enhance precision in forecasting prices of crude oil. In the current work, individual models such as Autoregressive Integrated Moving Average (ARIMA) and Support Vector Regression (SVR) along with hybrid models integrating SVR with K-Nearest Neighbors (KNN), Random Forest (RF), and Decision Tree (DT) are researched. Training as well as testing are conducted for the historical sets of Brent Crude, WTI Crude, and OPEC Basket data between the years 2019-2024. The findings establish that an individual SVR model outperforms ARIMA, with a score of 0.0511 MAE and 0.0683 RMSE to estimate the price of Brent Crude. In the instance of hybrid models, SVR+KNN signifies the minimum value of 0.3115 MAPE, outpacing SVR+RF as well as SVR+DT. Overall, SVR signals high prediction value for different time scales, thereby underscoring machine learning's capabilities to model intricate, non-linear financial time series.

Keywords— Crude oil price forecasting, hybrid ML, chaining models, SVR, DT, RF, KNN, ARIMA, financial time series, RMSE, MAPE, MAE.

I. INTRODUCTION

Crude oil, often called "black gold," is vital to the global economy, fuelling industries, transportation, and manufacturing. Accurate predictions of crude prices are crucial for decision-makers, including policymakers, businesses, and investors, given their significant economic implications [1]. However, predicting oil prices is challenging because of their inherent volatility, influenced by supply-demand dynamics, geopolitical risks, OPEC policies, and unexpected events such as natural disasters or conflicts.

For predicting crude oil prices, traditional time series models ARIMA and GARCH (Generalized Autoregressive Conditional Heteroskedasticity), have been widely used because of their effectiveness in analyzing patterns and volatility [2]. Although effective to some extent, these models often struggle to accurately represent the complex and nonlinear behavior of oil prices, particularly during times of significant volatility [3]. Furthermore, their heavy dependence on historical data and limited incorporation of external market factors reduces their adaptability to sudden shifts in market conditions.

In contrast, ML models offer advanced tools for analyzing large datasets and discover emerging trends [15]. By integrating historical price data with external market

indicators, these models can greatly enhance prediction accuracy. This helps to provide more reliable insights and support for better informed decision-making. This study investigates hybrid ML approaches, combining Techniques like SVR, KNN, RF, and DT, to create a reliable framework for predicting the price of crude oil.

II. RELATED WORK

The prediction of crude oil pricing has been a central focus of extensive research over the past several decades [1]. Traditional methods for forecasting include statistical models such as ARIMA, GARCH, have been extensively used, yet limited in capturing nonlinearities and market disruptions [2]. Despite their limitations, ARIMA and GARCH have remained popular due to their simplicity and interpretability [5].

TABLE 1. Overview of recent research on forecasting crude oil price.

Year of Publication of the research paper	Dataset Used	Model or Method Used	Drawbacks
[1] (2019)	The daily WTI and Brent spot prices of crude oil markets (1986-2018)	Multi-Class-Support Vector Machine (MCSVM), Non-dominated Sorting Genetic Algorithm II (NSGA-II)	Dependence on historical data, complexity, parameter sensitivity, limited price movement classes, and lack of real-time testing.
[2] (2021)	Oil price WTI and related news datasets	Open Domain Event Extraction (ODEE), Sentiment Analysis, Deep Learning Models	Limited exogenous variables, dependence on news quality, and model complexity.
[3] (2024)	Brent and WTI Dataset	LKDSR (LSTM-Kernel Density Support Regression), AdaBoost-LSTM, AdaBoost-GRU	Limited historical application, computational complexity, and homogeneity requirement.

[4] (2023)	Weekly WTI spot prices of crude oil (1998-2023)	Variational Mode Decomposition (VMD), Time-Series Imaging, Convolutional Neural Network (CNN), Bidirectional Gated Recurrent Unit Network (BGRU)	Dependence on univariate analysis, the necessity for future research to incorporate additional factors.
[5] (2019)	Brent (1970–2008) and WTI (1983–2010) crude oil price datasets	ARIMA, Kalman Filter (KF), Ensemble-Empirical-Mode-Decomposition (EEMD), Hidden Markov Models (HMM), LSSVM, GARCH	Complexity of implementation, focus on specific markets, and stochastic component analysis challenges.
[6] (2020)	Daily closing prices of WTI crude oil (1986-2019)	CEEMDAN, Multi-layer Gated Recurrent Unit (ML-GRU) networks	Mode mixing, residual noise, and model complexity.
[7] (2020)	Historical data of crude oil prices based on international markets	ANN, SVR, LSTM, Visibility Graph Algorithms, K-core Centrality, Hybrid AI Systems	Limited scope, data restriction, complexity of non-linearity, lack of validation with diverse datasets.
[8] (2020)	"Crude oil price data" collected from investing.com	Artificial Neural Network (ANN). Additionally, the Back-propagation learning algorithm is used to teach the ANN.	The inability to accurately forecast crude oil prices during abrupt and significant market fluctuations, underscoring the inherent unpredictability of financial markets.
[9] (2023)	WTI and Brent crude oil futures (2010-2023)	ICEEMDAN, Gate Recurrent Unit (GRU), Radial Basis Function Neural Network (RBF), Back Propagation Neural Network (BPNN), Multi-Objective Sparrow Search Algorithm (MOSSA)	Reliance on historical data, complexity, and lack of external factor consideration.
[10] (2023)	WTI crude oil prices (2018-2022)	Local Mean Decomposition (LMD), ARIMA, LSTM Networks	Univariate data limitation, stationarity requirement, model complexity.
[11] (2024)	WTI crude oil data, climate policy uncertainty index	Autoregressive (AR) model, CPU-related indicators (CCPU, ICPU)	Insignificant original CPU index, limited economic context, transaction cost assumptions.

[12] (2021)	WTI crude oil price dataset (1986-2020)	LSTM Neural Network, GM (11) Model, Hybrid Model Approach	Data limitations, model complexity, computational intensity.
[13] (2021)	Historical oil price data (1986-2017)	Multiple Regression, ARIMA Models, Combination of Forecasting Methods	Reliance on human expert knowledge, market volatility challenges.
[14] (2019)	Crude oil prices (2007-2019)	Hybrid Bayesian Network (BN), RF, SVM, MCMC, Neural Networks (NNET), GARCH	Overfitting, complexity, data sensitivity, limited scope of analysis.
[15] (2021)	WTI crude oil dataset (January 2000 to December 2017)	Elastic-Net, BMA, Spike and Slab LASSO, LSTM, RW, ARMA, ENN, ELM, WNN, GRNN	Complexity of models, data dependency, generalizability, computational resources

III. DATA COLLECTION

A. Data Sources

This research employed historical data from three key crude oil benchmarks: Brent Crude Oil Futures [20], WTI Crude Oil Futures [21], and the OPEC Basket Price [22], spanning 2019–2024. These data were taken from reliable platforms like investing.com and the OPEC website, providing a robust basis for modeling. The schedule encompassed daily, weekly, and monthly intervals to capture short- and long-term trends, with features such as opening price, closing price, daily high and low prices, and trading volumes.

- **Brent Crude Oil Futures (LCOX4):** Representing two-thirds of global oil pricing, this dataset includes 1,825 daily records, highlighting events like the April 2020 crash to \$19.33 per barrel during the COVID-19 pandemic and OPEC production cuts [20].
- **WTI Crude Oil Futures (CLX4):** A benchmark for the U.S. market, this dataset mirrors Brent's structure, with records of the historic drop to - \$37.63 per barrel in April 2020, reflecting oversupply and storage issues [21].
- **OPEC Basket Price:** This weighted average of crude blends from OPEC countries includes 1,825 records, showcasing price stabilization efforts post-2020, ranging from \$12 per barrel to highs above \$100 in 2022 [22].

B. Data Preprocessing

Data pre-processing is one of the crucial step for preparing the dataset by ensuring it is clean, consistent, and ready for modeling [7]. The pre-processed dataset provides a comprehensive view, including various technical indicators, prices, and market volatility indicators like the OVX (Oil

Volatility Index) and VIX (Volatility Index). Below is a thorough description of the data preprocessing steps:

a) Handling Missing Values

Since financial data is often prone to missing values due to non-trading days or data issues, we handled missing data using two primary methods. First, rows with null values were removed to ensure the dataset's integrity and avoid complications in analysis. Second, forward fill technique was applied to handle continuous missing values, ensuring that missing data points take the worth of the preceding data point. This is especially helpful in situations where missing data is temporary and can reasonably be assumed to follow the preceding trend [9].

b) Normalization

Crude oil price information and the associated indicators have varying scales. To prevent certain features, such as price, from dominating the learning process, we applied Min-Max Scaling. This transformation scaled all the features into a [0, 1] range, ensuring uniformity across the dataset. This step is especially crucial for algorithms such as SVR and KNN, which depend on distance-based metrics [2].

c) Feature Engineering

Feature engineering plays a vital part in improving the predictive ability of ML models. By creating new features from the existing data, we can discover patterns, trends, and signals in which the model can utilize to make more precise predictions.

C. Train-Test Split

Once the data preprocessing and feature engineering stages are finished, the next crucial step in developing a robust ML technique is dividing the dataset in testing and training sets. This ensures the model is assessed based on its capacity to generalize to new, unseen data, helping to prevent the model from overfitting or underfitting the training data [6]. The dataset was split into an **80:20 ratio**, where 80% of data was utilized to train the model, while the remainder 20% was reserved for testing [1].

IV. METHODOLOGY

The methodology section outlines the algorithms and models employed in this study for forecasting crude oil prices [10]. It begins with an analysis of standalone models, such as SVR and ARIMA, and it then moves on to explore hybrid models which combines various techniques to enhance predictive accuracy.

A. Standalone Models

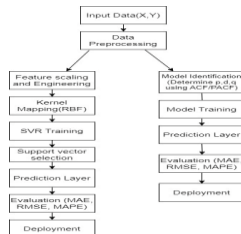


Fig. 1. Architecture of SVR and ARIMA Model

a) SVR

SVR is a powerful ML technique often applied in regression problems, particularly when the relationship between input features and output variables is non-linear [7]. SVR aims to identify a hyperplane that effectively divides data points in a high-dimensional space while allowing for a defined margin of error [10]. For this analysis, SVR is utilized for predicting the price of crude oil by mapping input data into a feature space with higher-dimensions using either linear or non-linear mapping. This transformation is achieved using kernel functions, with the RBF (Radial Basis Function) kernel proving particularly effective in capturing intricate patterns in time-series data. The predictive function for SVR is given by:

$$f(x) = \sum_{i=1}^N (a_i - a_i^*) K(x_i, x) + b \quad \dots (1)$$

From equation (1), x represents the input vector for prediction, and x_i denotes the support vectors from the training data. The coefficients a_i and a_i^* are Lagrange multipliers obtained during optimization. The kernel function $K(x_i, x)$ measures the similarity between x_i and x , allowing SVR to handle non-linear relationships. Finally, b represents the bias term that is determined during the optimization process, completing the prediction mode.

b) ARIMA

ARIMA is a commonly utilized statistical method used to forecast time series data [6]. It combines autoregression, differencing, and moving averages to model and predict future values based on historical data. In the ARIMA model, the equation is given by:

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} \quad \dots (2)$$

From given equation (2), x_t is the observed value at time t , $\phi_1, \phi_2, \dots, \phi_p$ are the autoregressive (AR) coefficients, and $\theta_1, \theta_2, \dots, \theta_p$ are the moving average (MA) coefficients. ϵ_t represents white noise at time t , while p , d , and q are the orders of the AR terms, differencing (for stationarity), and MA terms, respectively.

B. Proposed Hybrid Models

This research seeks to enhance the precision of crude oil price prediction by using hybrid models which combine various algorithms. By combining various methodologies, these are designed to effectively extract linear and nonlinear relationships in the data, which leads to their enhanced predictive accuracy. SVR is very accurate in dealing with nonlinearity but sensitive to hyperparameters and noise. Tree based regressors like RF, DT are better suited to deal with hierarchical structures and deal with data irregularities. KNN regressors, while being simple, are non-parametric and can readily accommodate local data patterns.

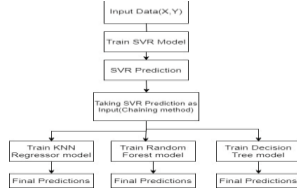


Fig. 2. Architecture of SVR+KNN, SVR+RF and SVR+DT Hybrid Model

a) SVR + KNN Hybrid Model

The SVR + KNN hybrid model integrates the global trend-capturing capability of SVR with the local pattern-detection advantage of the KNN algorithm, offering a complementary approach to modelling crude oil rates.

Equation given by:

- **SVR Prediction Stage:** For an input x , SVR predicts an intermediate output y_{SVR} using equation (1).
- **Feature Augmentation:** The SVR output y_{SVR} is used as an additional feature for the KNN model. The augmented input vector becomes:

$$x' = [x, y_{SVR}]$$

- **KNN Prediction Stage:** The KNN model predicts the final output y using the augmented input vector x' :

$$y = \frac{1}{k} \sum_{i \in N(x')} y_i \quad \dots (3)$$

By using equation (3), $N(x')$ represents the collection of indices of the KNN for the augmented input vector x' , which includes both the original features and the SVR output. The values y_i correspond to the observed values of these neighbors in the augmented feature space. These neighbors are used by the KNN algorithm to compute the final prediction as the average of the observed values y_i from the closest neighbors in the feature space.

b) SVR + RF Hybrid Model

The SVR + RF hybrid model combines the non-linear regression capability of SVR with the ensemble learning strength of RF, providing a reliable framework for prediction of crude oil prices.

Equation given by:

- **SVR Prediction Stage:** For an input x , SVR predicts an intermediate output y_{SVR} using equation (1).
- **Feature Augmentation:** The SVR output y_{SVR} is used as an additional feature for the RF model. The augmented input vector becomes:

$$x' = [x, y_{SVR}]$$

- **RF Prediction Stage:** The RF model uses the augmented input vector x' to estimate the final output y by aggregating predictions from multiple decision trees:

$$y = \frac{1}{T} \sum_{t=1}^T Tree_t(x') \quad \dots (4)$$

From the equation (4), T represents the quantity of trees in the forest, $Tree_t(x')$ denotes the prediction made by the t -th DT for the augmented input x' . Each tree in the forest independently makes a forecasting using the augmented feature vector, and the final output is obtained by averaging the predictions from all T trees.

c) SVR + DT Hybrid Model

SVR + DT hybrid model adopts a hierarchical approach, leveraging the global trend-detection strengths of SVR and the rule-based, interpretative capabilities of DT.

Equation given by:

- **SVR Prediction Stage:** For an input x , SVR predicts an intermediate output y_{SVR} using equation (1).
- **Feature Augmentation:** The SVR output y_{SVR} is used as an additional feature for the DT model. The augmented input vector becomes:

$$x' = [x, y_{SVR}]$$

- **DT Prediction Stage:** The DT model uses the augmented input vector x' to estimate the final output y based on the decision rules derived from the training data:

$$y = DecisionTree(x') \quad \dots (5)$$

By using equation (5), the augmented feature vector x' includes both the original features and the SVR output. The DT then splits the feature space is divided into regions according to these features and assigns a prediction by averaging or taking most of the training samples in each region.

V. RESULTS AND ANALYSIS

This section showcases the outcomes of the predictive modeling performed to estimate crude oil prices, comparing standalone and hybrid models using various evaluation metrics. The models evaluated include SVR, ARIMA, and hybrid models combining SVR with RF, KNN, and DT.

A. Performance Metrics

a) Mean Absolute Error (MAE)

MAE calculates the average size of the errors between predicted and actual values, providing a straightforward interpretation of model accuracy. It as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad \dots (6)$$

From equation (6), n represents the number of data points (or observations). y_i denotes the actual value, whereas $\hat{y}_i$ is the predicted value for the i -th observation. The term $|y_i - \hat{y}_i|$ represents the absolute difference between the actual vs predicted values, ensuring that both under-predictions and over-predictions are treated equally by calculating the absolute value of the difference.

b) Root Mean Squared Error (RMSE)

RMSE is the square root of the mean of the squared differences between the observed and forecasted values. Due to the squared term, it is more sensitive to outliers than MAE.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad \dots(7)$$

From equation (7), n represents the number of data points (or observations). Y_i denotes the actual value for the i -th observation, while $\hat{y}_i$ is the predicted value for the same observation. The term $(y_i - \hat{y}_i)^2$ represents the squared difference between the observed and forecasted values, which penalizes larger errors more significantly.

c) Mean Absolute Percentage Error (MAPE)

It is used as a magnitude-independent prediction accuracy measure. Whereas MAE and RMSE are data magnitude-sensitive, MAPE presents error as a percentage of actual values, facilitating easier interpretation and model performance comparison across datasets such as Brent, WTI, and OPEC having varying value ranges. MAPE tends to favor models that underestimate rather than overestimate. It calculates the percentage difference between predicted and actual values:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \times 100 \quad \dots(8)$$

From equation (8), Y_i represents the actual value, while $|y_i - \hat{y}_i|$ denotes the predicted value. The result is then multiplied by 100 to express the error as a percentage.

B. Model Performance

We evaluated both standalone models (SVR, ARIMA) and hybrid models (SVR combined with RF, KNN, and DT) based on these metrics across daily, weekly, and monthly periods.

Table 2. Error values of crude oil closing price on Daily Brent Crude Oil data set

Model	MAE	RMSE	MAPE
SVR (Standalone)	0.0511	0.0683	0.3115
ARIMA (Standalone)	4.7838	5.6758	5.8371
SVR + RF (Proposed)	1.3065	1.6937	0.0156
SVR + KNN (Proposed)	1.2282	1.6029	0.0147
SVR + DT (Proposed)	1.4982	1.9155	0.0179

The standalone SVR model outperforms others, demonstrating the lowest MAE of 0.0511 and RMSE of 0.0683, indicating superior accuracy and a better fit for daily data. However, its MAPE of 0.3115 suggests slightly higher percentage deviations than other metrics. Among the hybrid models, SVR + KNN delivers the better outcome, even though it falls short of the standalone SVR in both MAE and RMSE. This underscores SVR's effectiveness in minimizing prediction errors and capturing data patterns.

Table 3. Error values of crude oil closing price on Weekly Brent Crude Oil data set

Model	MAE	RMSE	MAPE
SVR (Standalone)	0.1204	0.1614	0.6873
ARIMA (Standalone)	11.1788	12.2552	13.9270
SVR + RF (Proposed)	3.6394	4.4398	0.0443
SVR + KNN (Proposed)	3.1756	3.9626	0.0389
SVR + DT (Proposed)	4.3433	5.2118	0.0527

The standalone SVR model consistently exhibits the lowest errors, with RMSE of 0.1614 and MAE of 0.1204, indicating its superior ability to generate predictions closely aligned with actual values. In contrast, ARIMA performs significantly worse. Among the hybrid approaches, the SVR + KNN combination remains the most effective, though it still fails to surpass the standalone SVR in predictive accuracy.

Table 4. Error values of crude oil closing price on Monthly Brent Crude Oil data set

Model	MAE	RMSE	MAPE
SVR (Standalone)	0.2010	0.2345	8.9570
ARIMA (Standalone)	11.3469	12.2211	13.8375
SVR + RF (Proposed)	6.1392	7.5327	0.0740
SVR + KNN (Proposed)	5.6012	6.2176	0.0677
SVR + DT (Proposed)	7.4683	8.6254	0.0909

The standalone SVR model remains the top-performing approach, achieving the lowest error metrics, with MAE about 0.2010 and RMSE about 0.2345. While its MAPE is relatively higher at 8.9570, it is still significantly better than ARIMA, which exhibits much larger errors. Among the hybrid models, SVR + KNN demonstrates the strongest performance, showcasing robustness, but it still falls short of matching the accuracy of the standalone SVR model across all metrics.

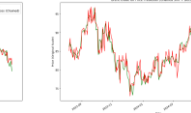
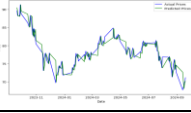
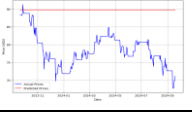
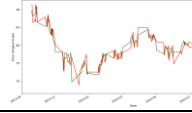
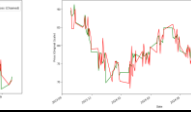
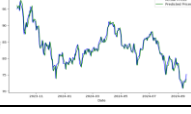
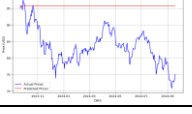
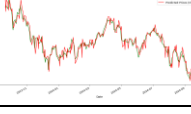
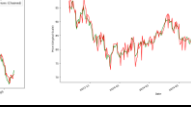
We compared the performance of our proposed model with an existing work that is MLR-SVR model proposed by Xinrui Liu [23]. Our proposed model SVR+RF and SVR+KNN exhibit relatively lower MAPE as given below:

Table 5. MAPE comparison among SVR+RF, SVR+KNN and MLR-SVR

Model	MAPE
SVR (Standalone)	0.3115
ARIMA (Standalone)	5.8371
SVR + RF (Proposed)	0.0156
SVR + KNN (Proposed)	0.0147
SVR + DT (Proposed)	0.0179
MLR-SVR [23]	0.0162

C. Graph Visualization

Table 6. Actual Vs Predicted crude oil closing price on Daily data set for three datasets (Brent Crude Oil, WTI Crude Oil, OPEC Crude Oil) using five models

Models	SVR	ARIMA	SVR + RF	SVR + KNN	SVR + DT
Dataset					
Brent Crude Oil (Daily)					
WTI Crude Oil (Daily)					
OPEC Crude Oil (Daily)					

D. Comparative Analysis

The comparative analysis highlights the following key insights:

- **Model Accuracy:** The standalone SVR model emerges as the most accurate technique for the price of crude oil forecasting, with significantly lower prediction errors compared to both ARIMA and hybrid models.
- **Hybrid Model's Robustness:** While hybrid models, especially those combining SVR and KNN, demonstrate better performance than ARIMA, they do not exceed the accuracy of standalone SVR. This indicates that although hybrid models can capture complex patterns, they may also introduce unnecessary complexity without significantly enhancing the results.
- **Performance Consistency:** SVR exhibits consistent performance across different time frames (daily, weekly, and monthly), indicating its reliability for various forecasting horizons.

VI. CONCLUSION

This research compared the standalone and hybrid methods for predicting the price of crude oil at daily, weekly, and monthly horizons. Standalone SVR model outperformed conventional models such as ARIMA and hybrid models in all the measures such as MAE, RMSE, and MAPE is most notably demonstrated by a daily MAE of 0.0511 and RMSE of 0.0683. While hybrid models such as SVR + KNN performed comparatively better among blended algorithms, they were less accurate than SVR. These findings bear witness to the dependability and

strength of SVR in holding the instability of crude oil price in check and reaffirm the excellence of sophisticated ML algorithms over conventional statistical models in time-series forecasting.

Future research should explore integrating SVR with advanced feature engineering and real-time data streams to further enhance performance. Additionally, incorporating deep learning methods and analyzing external elements like geopolitical events and macroeconomic indicators, could provide a more comprehensive framework for improving forecasting in this dynamic market.

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